DIDULANI ACHARIGE
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EDUCATIONAL HISTORY

Ph.D. in Mechanical and Materials Engineering (Direct Ph.D. program)
Western University, London, Ontario, Canada (Cumulative GPA – 91.125/100) **2019 – 2024**

- Thesis – "**Deep Learning for inverse design of nanophotonic structures**."
- *Advisor - Dr. Eric Johlin*
  Led research initiative focused on enhancing algorithmic design of 2D nanophotonic structures for solar cell applications, 3D nanophotonic structures for quantum lens application, along with more complex color splitter applications for standard cameras. Both machine learning and deep learning techniques were used for the design and optimization process. Achieved significant improvements in performances, contributing to advancements in photonic technology.

B.Sc. in Engineering Physics with Honors
University of Colombo, Sri Lanka (GPA – 3.53/4.0) – First Class Honors **2015 – 2019**

Subsidiary subjects: Statistics, Computer Science, Applied Mathematics

- Thesis – "**Development of a combinatorial optical property measuring system for classification / quantification of liquids**."
- *Advisor - Dr. Hiran Jayaweera*
  Designed and developed a combinatorial optical property measurement system to classify and quantify liquids with precision. The system utilized LEDs of varying wavelengths and photodetectors to measure key optical properties, including transmittance, scattering, and refraction. Implemented advanced data compression and multivariate analysis techniques, leveraging machine learning methods to enhance accuracy. A deep neural network was developed and deployed to effectively classify and quantify liquid samples, demonstrating the system's robust analytical capabilities.

EMPLOYMENT HISTORY

Postdoctoral Research Associate **2025 – Current**
Western University, London, Ontario, Mechanical and Materials Engineering Department

- Further examine Generative AI techniques for the design, optimization of 2 layer and 3-layer color splitter structures to replace traditional color filters in standard cameras, achieving a significant improvement in light transmission and reduced absorption.
- Validated the AI model by comparing their generated designs with the performance of actual fabricated color splitter structures.
- Two photon lithography nanofabrication system.

Post Graduate Research Assistant – Nanophotonic Energy Materials Lab **2024 – 2025**
Western University, London, Ontario, Mechanical and Materials Engineering Department

- Applied Generative AI for the design, optimization of color splitter structures to replace traditional color filters in standard cameras, achieving a significant improvement in light transmission and reduced absorption.

Research Assistant – Nanophotonic Energy Materials Lab **2019 – 2024**
Western University, London, Ontario, Mechanical and Materials Engineering Department

- Applied Generative AI for the design and optimization of 2D nanophotonic structures to enhance the light absorption and increase the efficiency of solar cells.
- Deep learning methodologies (Tandem Neural Network) were employed for the design and optimization of 3D nanophotonic quantum lenses, to enhance the photon collection in quantum computers.
- Nano lens structures were designed and optimized using deep learning techniques (Convolutional Neural Network and ResNet50) to serve as single-photon emitters for quantum dot applications.
- Implemented a deep learning algorithm to remove bird contamination from wind profile radar data for data collected from Federal Office of Meteorology and Climatology in Payerne Switzerland to enhance the weather prediction accuracy.
- Conducted simulations and model optimizations to preprocess and analyze large volumes of structures and unstructured data, identifying trends and patterns to build accurate predictive models.
- Implemented ensemble deep learning modeling techniques to combine multiple models, improving

prediction accuracy and efficiency.

- Developed image processing, recognition, and computer vision solutions, gaining hands-on experience in these areas.

Teaching Assistant **2020 – 2024**
Western University, London, Ontario, Mechanical and Materials Engineering Department

- Mentored students in applying mathematical models and statistical techniques and conducted lab sessions in courses including Engineering Experimentation, Control Systems, Materials Selection, and Energy Conversion.
- Assisted in the development of course materials and assessments, ensuring students gained practical experience in data analysis, system optimization, and engineering experimentation.

Assistant Lecturer **2019 Feb – 2019 June**
University of Colombo, Sri Lanka, Department of Physics

- Taught coursers, conducted lab sessions, tutorials and mentored students in experiments and programming, and collaborated with peers to optimize teaching strategies enhancing student outcomes.

PUBLICATIONS Google scholar

- D Acharige, E Johlin. "Machine learning in interpolation and extrapolation for nanophotonic inverse design", *ACS omega*, **7,** (37), 33537-33547
- D Acharige, E Johlin. "Machine learning and artificial neural networks for improved algorithmic design of nanophotonic structures.", Canadian Conference on AI
- D Acharige, E Johlin. Hybrid Deep Learning for Design of Nanophotonic Lenses, *IOP Nano Express, **5,** 035014*
- D Acharige, E Johlin, "Generative Tandem Neural Network for optimization of photonic structures", *Physica Scripta, **4,** 046006*

MANUSCRIPTS IN PREPERATION

- D Acharige, E Johlin. "Experimental Validation of Deep Learning Designed Nanophotonic Color Splitter Structures." (In preparation)

ORAL AND POSTER PRESENTATIONS

- "AI for nanophotonics", 120 second research sprint, Postdoctoral Research Forum, Western University, London, Ontario, June 5, 2025
- "Artificial Neural Networks for the design and optimization of color splitter structures" , *Photonic North Conference*, Vancouver, British Columbia, May 29th, 2024
- "Generative Tandem Neural Network for optimization of photonic structures", Western University, London, Ontario, *SOGS Western Research Forum*, March 15th, 2024
- "Artificial Neural Networks to design and optimize color splitters", *Department seminar,* Mechanical and Material Engineering, Western University, November 22nd, 2023
- "Machine Learning and Artificial Neural Networks for improved algorithmic design of nanophotonics" , *Department seminar,* Mechanical and Materials Engineering, Western University, October 20th, 2022
- "Comparison of different machine learning and artificial neural network techniques for nanophotonic structure generation", *Department seminar,* Mechanical and Material Engineering, Western University, October 25th, 2021
- "Machine Leaning and Artificial Neural Networks for improved algorithmic design of nanophotonic structures" Western University, London, Ontario, CAMBR Day, June, 17th 2021
- "Machine Learning and Artificial Neural Networks for improved algorithmic design of nanophotonic structures", Graduate Student Symposium, Vancouver, British Columbia, *Canadian AI*, May, 25th 2021

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HONOURS AND AWARDS

Western Graduate Research Scholarship
Western University, London, Ontario, Canada **2019 – 2024**

MITAC Research Training Award
MITAC, Canada **2020 Jan – 2020 April**

Mahapola Higher Education Scholarship
Government of Sri Lanka **2015 – 2019**

- Full undergraduate scholarship for 4 years based on the academic performance in the university admission examination, GCE Advanced level)

PROJECT BASED EXPERIENCE

- End effector of a pick and place robot arm as an innovative solution for robotic and effectors using SolidWorks.
- Implemented generative neural network models (Convolutional Neural Network, Deep Neural Network and Conditional Generative Adversarial Network model and Principal Component Analysis) to generate facial data related to different age parameters for facial recognition system.
- AVR programming was used for embedded systems, i.e., designed electronics and programming of the LED cube; constructed electronics and a programming of the line following robot, path following robot, and navigation-based robots. Pick-and-place robot arm construction and programming using C language.
- Designed and developed a prototype of a mini drill press machine by using AutoCAD and Solid work software. It was constructed to use as a PCB circuit board drill machine by acquiring workshop experience. Under this project, lathe machines, milling machines, drilling machines, saw machines, & other mechanical workshop tools were utilized for the implementation.

PROFESSIONAL SKILLS

- **Artificial Intelligence**: Machine Learning, Deep Learning, Generative AI, Transfer Learning, Predictive modeling, Computer Vision, Statistics
- **Programming**: Python (TensorFlow, Keras, scikit-learn, Pandas, NumPy, OpenCV), MATLAB, C, VHDL, Arduino, LaTeX
- **Frameworks & Libraries**: TensorFlow, Keras
- **Technical Software**: Meep (FDTD), SolidWorks, AutoCAD, LabVIEW, Simulink, MultiSim, ModelSim, MATLAB, Inkscape
- **Machinery experience**: Lathe, Milling, Drilling, Saw & other mechanical workshop tools

VOLUNTEER AND LEADERSHIP EXPERIENCE

- Organized Physics workshops for high school teachers; and guided, delegated, and motivated fellow undergraduate students in their tasks
- Assistant secretary of Physics Society, University of Colombo
- Member of the university official media team (FOS media), University of Colombo
- Teacher of the high school mathematics supportive classes for children from unprivileged families